Site-to-site IPsec VPN with SonicWall TZ370

Company ABC Seattle to Denver VPN

For Company ABC, I selected a SonicWall TZ370 for both the Seattle and Denver offices and designed a site to site IPsec VPN using IKEv2. The goal was to give employees secure access to approved resources at the other branch over the internet. This scenario assumes modest traffic requirements and static public IP addresses. The walkthrough describes the configuration for the scenario; the final section covers validation to perform before deployment.

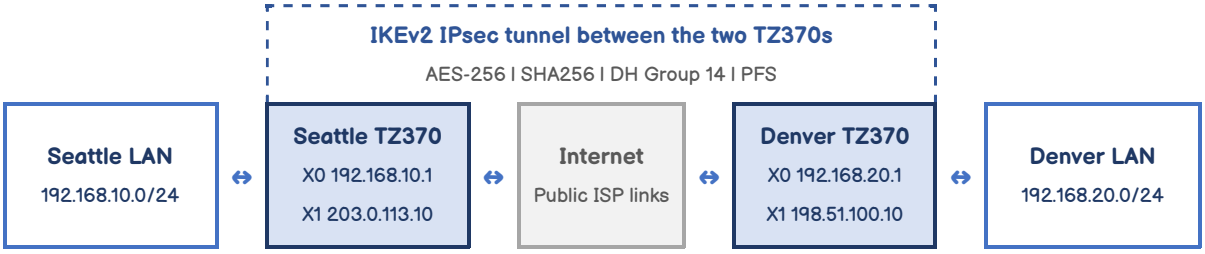

Figure 1. Seattle to Denver site to site VPN topology. Original diagram created for this post.

1 Establish the addressing plan

I assigned the following example addresses to keep the two branch networks distinct:

Setting	Seattle	Denver
WAN address	203.0.113.10	198.51.100.10
WAN subnet mask	255.255.255.0	255.255.255.0
ISP gateway	203.0.113.1	198.51.100.1
LAN network	192.168.10.0/24	192.168.20.0/24
LAN gateway (X0)	192.168.10.1	192.168.20.1

The WAN addresses are documentation examples. An actual deployment would use public addresses, subnet masks, and gateways supplied by each ISP. I chose different private networks so the branches would not have overlapping addresses.

2 Complete the initial firewall setup

I started with SonicWall's setup guidance, connecting an administrator's computer to the X0 LAN port and opening **https://192.168.168.168**. From there, I signed in with the initial credentials, changed the administrator password, completed the setup wizard, and registered each appliance with MySonicWall. A supported firmware release and a configuration backup belong in the preparation process before deployment. The menu paths below follow SonicOS 7.

3 Configure the LAN and WAN interfaces

Next, I configured X0 with each branch's LAN gateway and X1 with its WAN address, subnet mask, and ISP gateway under **Network > System > Interfaces**. Seattle used **192.168.10.1** on X0, and Denver used **192.168.20.1**. Each branch firewall served as the default gateway for its local computers.

I restricted HTTPS firewall administration to approved administrator computers. Before moving on to the

VPN, internet connectivity needs to be confirmed at both offices so an ISP or interface issue does not get mistaken for a tunnel problem.

4 Create the network address objects

Under **Object > Match Objects > Addresses**, I created a Network object for the remote branch and assigned it to the VPN zone. On Seattle's firewall, **Denver_LAN** represented **192.168.20.0** with mask **255.255.255.0**. On Denver's firewall, **Seattle_LAN** represented **192.168.10.0** with the same mask. The existing **X0 Subnet** object on each firewall identified its local network.

5 Create Seattle's VPN policy

To build Seattle's policy, I opened **Network > IPsec VPN > Rules and Settings**, enabled VPN, and selected **Add**. I chose **Site to Site** as the policy type and **IKE using Preshared Secret** as the authentication method.

I named the policy **Seattle_to_Denver** and entered **198.51.100.10** as the IPsec Primary Gateway. Authentication required the same long, randomly generated shared secret on both firewalls, entered securely. On the **Network** tab, I selected Seattle's **X0 Subnet** as the local network and **Denver_LAN** as the destination network.

6 Configure matching security proposals

On the **Proposals** tab, I selected **IKEv2 Mode** and configured the following security settings:

Setting	IKE (Phase 1)	IPsec (Phase 2)
Protocol	IKEv2	ESP
Encryption	AES-256	AES-256
Authentication	SHA256	SHA256
Diffie-Hellman	Group 14	PFS with Group 14
Lifetime	28,800 seconds	3,600 seconds

I matched these proposals on both firewalls so the peers could negotiate the connection. The settings reflect the security choices for this scenario rather than simply accepting every default.

7 Configure Denver's matching policy

I repeated the policy setup on Denver's firewall, naming it **Denver_to_Seattle**. This time, the remote gateway was Seattle's WAN address, **203.0.113.10**. Denver's X0 Subnet became the local network, and Seattle_LAN became the destination network.

Policy setting	Seattle firewall	Denver firewall
Policy name	**Seattle_to_Denver**	**Denver_to_Seattle**
Primary gateway	**198.51.100.10**	**203.0.113.10**
Local network	**X0 Subnet**	**X0 Subnet**
Destination network	**Denver_LAN**	**Seattle_LAN**

The Denver policy used the same shared secret and security proposals as Seattle. Under **Advanced**, I

bound both policies to the WAN zone and left unnecessary broadcast and firewall management options disabled. SonicWall's IKEv2 guidance recommends Keep Alive on at least one side; where the firmware exposes that option in IKEv2 mode, Seattle is the designated side.

8 Control which traffic can cross the tunnel

To control access across the tunnel, I selected **Suppress automatic Access Rules creation for VPN Policy**. I then defined explicit rules under **Policy > Rules and Policies > Access Rules** for approved LAN to VPN and VPN to LAN traffic on each firewall.

For the example HTTPS connection, I allowed an approved Seattle workstation to reach a Denver application server and added the corresponding receiving rule in Denver. Connections initiated from Denver require their own approved rules. Temporary ICMP rules can support testing. I left **Apply NAT Policies** disabled because the branch networks do not overlap.

9 Validate and document the connection

Validation starts by generating traffic between approved test computers and checking for an active tunnel. The next checks are to test the actual application, confirm that traffic counters increase, and verify that an unapproved connection is blocked. These checks establish whether the configuration works as intended; no measured test results are included in this walkthrough.

If the tunnel does not establish, the first items to check are the peer addresses, shared secret, proposals, and network selections. If the tunnel connects but applications fail, troubleshooting moves to access rules, endpoint firewalls, gateways, and DNS. Once validation is complete, the final configurations and test results should be saved for future troubleshooting.

> For me, a successful setup means more than seeing an active tunnel. Seattle and Denver should be able to use the resources they need while access remains limited to what Company ABC has approved.

References

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Figure 1 is my own diagram based on my addressing plan. No external illustrations were used.